

Microservice Architecture – Task 1

Link to GitHub

https://github.com/Romko-studik/micro_basics/tree/micro_basics

Architecture

The system includes three services:

- **facade-service:** Handles POST/GET HTTP requests from the client
- **logging-service:** Stores messages with deduplication and retry support
- **messages-service:** Returns a static message over HTTP

Running the Services

Run in separate terminals:

```
# Terminal 1
cd logging-service
uvicorn main:app --port 5000
```

```
# Terminal 2
cd facade-service
uvicorn main:app --port 6000
```

```
# Terminal 3
cd messages-service
uvicorn main:app --port 7000
```

Tests:

```
romko@fedora:~/Documents/UNI/APZ/micro_basics$ /micr-X POST http://localhost:5000/post -H "Content-Type: application/json" -d '{"msg": "Привіт, світ!"}'
romko@fedora:~/Documents/UNI/APZ/micro_basics$

{"detail": "Not Found"}romko@fedora:~/Documents/UNI/APZ/micro_basics$ curl http://localhost:5000/get
{"response": {"messages": [{"Привіт, світ!}]}\\n---\\n{"message": "not implemented yet!}"}romko@fedora:~/Documents/UNI/APZ/micro_basics$

{"detail": "Not Found"}romko@fedora:~/Documents/UNI/APZ/micro_basics$ curl http://localhost:6000/all
{"messages": [{"Привіт, світ!}]}romko@fedora:~/Documents/UNI/APZ/micro_basics$
```

Now we turn off logging service for some time to test retry

```
INFO:      Unicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
ERROR:root:Attempt 1 failed with connection error: All connection attempts failed
ERROR:root:Attempt 2 failed with connection error: All connection attempts failed
ERROR:root:Attempt 3 failed with connection error: All connection attempts failed
ERROR:root:Attempt 4 failed with connection error: All connection attempts failed
ERROR:root:Attempt 5 failed with connection error: All connection attempts failed
ERROR:root:Attempt 6 failed with connection error: All connection attempts failed
INFO:httpx:HTTP Request: POST http://localhost:6000/log "HTTP/1.1 200 OK"
INFO:root:Successfully sent message 31e3b127-4952-4650-9ee5-99649ffacc8f on attempt 7
INFO:      127.0.0.1:48876 - "POST /post HTTP/1.1" 200 OK
```

We can see that facade service retries a couple of times until logging service is back online.